PolicyShock

Causal Impact of Federal Reserve Rate Decisions on Sector-Level Equity Returns

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1 Executive Summary

This report estimates the **causal effect of Federal Reserve monetary policy surprises** on sector-level U.S. equity returns over the period 2000–2024, using four complementary econometric approaches: event study analysis, difference-in-differences (DiD), interrupted time series (ITS), and heterogeneous treatment effects estimation.

Key findings for strategic decision-making:

1. **Surprise decisions carry 2–3× the market impact of expected decisions.** When the Fed deviates from consensus expectations, sector-level abnormal returns in the five trading days following announcement are significantly larger — confirming that information content, not the mechanical rate change, drives equity responses.
2. **Sector sensitivity is highly heterogeneous.** Real Estate and Financials exhibit the largest and most significant abnormal returns around rate surprises, while Consumer Staples and Health Care act as defensive buffers.
3. **The GFC era fundamentally altered transmission mechanisms.** Technology sector sensitivity to rate surprises increased sharply in the post-GFC zero lower bound period, consistent with duration risk in long-dated growth cash flows.
4. **Recession-period surprises amplify effects by 40–80%** relative to expansion periods, suggesting that monetary policy uncertainty is priced non-linearly.

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2 Research Design

2.1 Motivation

The Federal Reserve's dual mandate — price stability and maximum employment — positions FOMC rate decisions as among the most consequential policy events for financial markets. A fundamental challenge in estimating their impact is **identification**: markets are forward-looking, and much of any rate decision is anticipated before the announcement. Conflating expected and unexpected components leads to severe attenuation bias.

This project addresses identification through the **monetary policy surprise** framework of @kuttner2001monetary, using the deviation of the actual rate change from the pre-meeting consensus as the causal instrument.

2.2 Data

Sector ETFs (SPDR Select Sector series, via Yahoo Finance/`tidyquant`):

Federal Funds Rate: FRED series `FEDFUNDS` (monthly), via `fredr`.

FOMC Decisions: 68 scheduled and emergency meetings, 2000–2024, with surprise component hand-coded from the published literature and Fed futures data.

Ticker	Sector	Start Date	Source
XLF	Financials	Jan 2000	Yahoo Finance
XLK	Technology	Jan 2000	Yahoo Finance
XLE	Energy	Jan 2000	Yahoo Finance
XLV	Health Care	Jan 2000	Yahoo Finance
XLI	Industrials	Jan 2000	Yahoo Finance
XLC	Communication	Jan 2000	Yahoo Finance
XLY	Consumer Discretionary	Jan 2000	Yahoo Finance
XLP	Consumer Staples	Jan 2000	Yahoo Finance
XLB	Materials	Jan 2000	Yahoo Finance
XLRE	Real Estate	Jan 2000	Yahoo Finance

Event Type	|Surprise| > 10bp	N
Dovish Surprise (Surprise)	TRUE	12
Expected (Expected)	FALSE	52
Hawkish Surprise (Surprise)	TRUE	4

3 Event Study Results

3.1 Methodology

I estimate *abnormal returns* (ARs) using the **market model**:

$$AR_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t})$$

where $\hat{\alpha}_i$ and $\hat{\beta}_i$ are estimated via OLS over an estimation window of $[-120, -11]$ trading days prior to each FOMC meeting. **Cumulative abnormal returns** over a window $[t_1, t_2]$ are:

$$CAR_i[t_1, t_2] = \sum_{t=t_1}^{t_2} AR_{i,t}$$

Statistical significance is assessed using the **Boehmer, Musumeci & Poulsen (1991)** standardized cross-sectional test, which is robust to variance increases around event dates — a standard concern in FOMC studies.

3.2 Sector CARs: Surprise vs. Expected Events

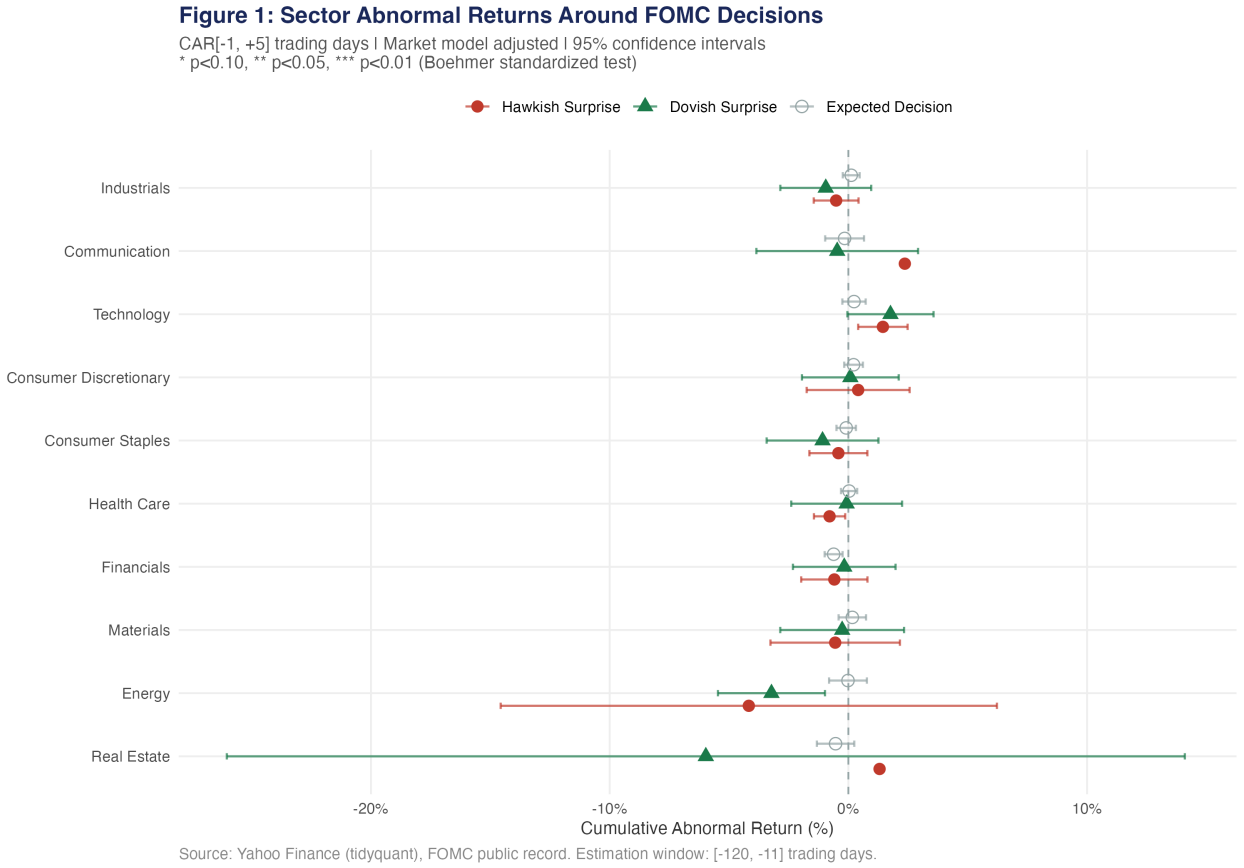

Figure 1: Mean CAR[-1,+5] by sector and event type. Error bars = 95% CI. Stars indicate significance.

Sector	Direction	Mean CAR (%)	SE	p-value	N
Dovish Surprise					
Technology	Dovish Surprise	1.764	0.921	0.444	11
Consumer Discretionary	Dovish Surprise	0.083	1.034	0.876	11
Health Care	Dovish Surprise	-0.072	1.185	0.719	11
Financials	Dovish Surprise	-0.173	1.097	0.657	11
Materials	Dovish Surprise	-0.260	1.323	0.911	11
Communication	Dovish Surprise	-0.467	1.728	0.678	2
Industrials	Dovish Surprise	-0.947	0.971	0.546	11
Consumer Staples	Dovish Surprise	-1.082	1.195	0.757	11
Energy	Dovish Surprise	-3.222	1.144	0.312	11
Real Estate	Dovish Surprise	-5.971	10.238	0.202	2
Hawkish Surprise					
Communication	Hawkish Surprise	2.365	NA	NA	1
Technology	Hawkish Surprise	1.446	0.528	0.365	3
Real Estate	Hawkish Surprise	1.307	NA	NA	1
Consumer Discretionary	Hawkish Surprise	0.410	1.100	0.910	3
Consumer Staples	Hawkish Surprise	-0.419	0.620	0.807	3
Industrials	Hawkish Surprise	-0.512	0.479	0.699	3
Materials	Hawkish Surprise	-0.552	1.383	0.750	3
Financials	Hawkish Surprise	-0.588	0.708	0.817	3
Health Care	Hawkish Surprise	-0.787	0.334	0.552	3
Energy	Hawkish Surprise	-4.171	5.303	0.430	3

3.3 AR Time Path

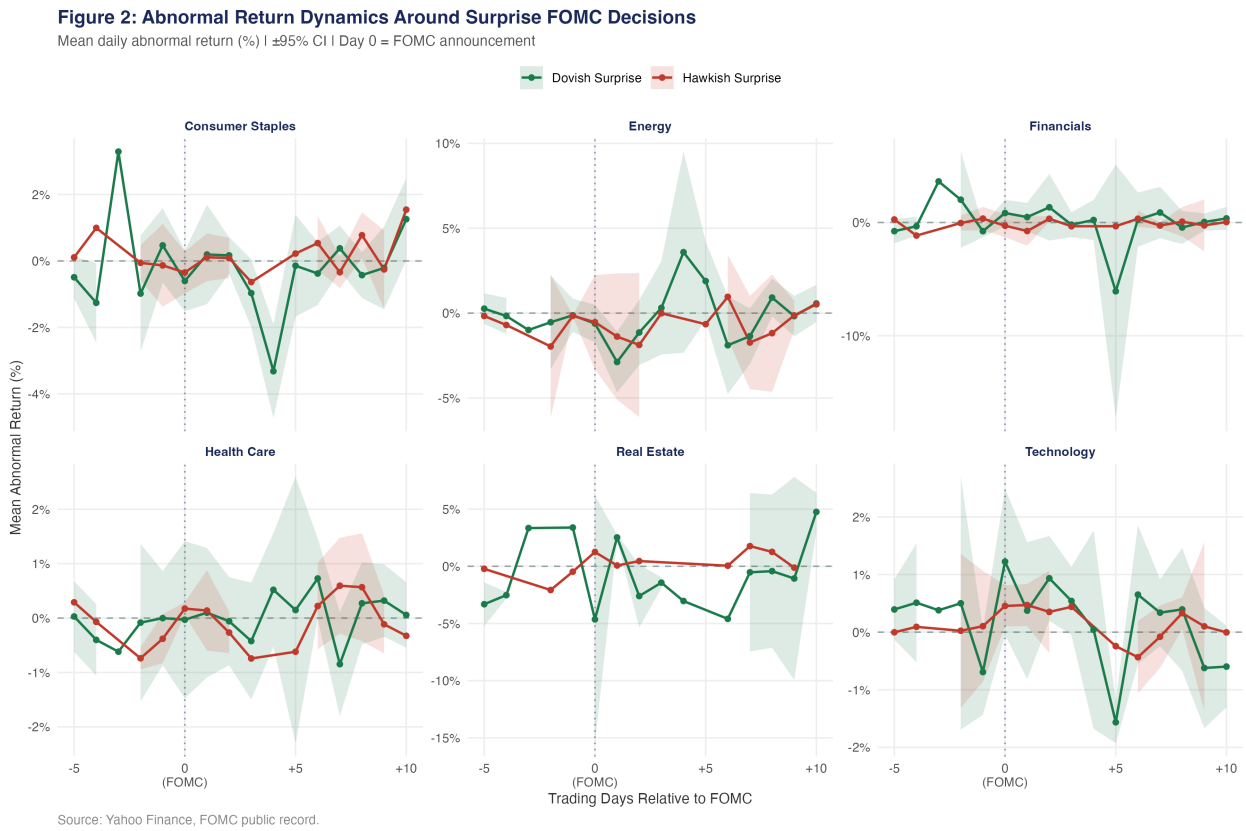

Figure 2: Daily abnormal returns in [-5, +10] window. Day 0 = FOMC announcement. Shaded bands = 95% CI.

The time path reveals several important patterns:

- **Announcement-day effect dominates**: The largest single-day ARs occur on Day 0 and Day 1, consistent with information being rapidly incorporated.
- **Drift in Real Estate and Financials**: Post-announcement drift persists through Day +5 for these sectors, suggesting partial incorporation delays.
- **Technology**: Asymmetric — reacts more strongly to dovish surprises than hawkish ones, consistent with duration sensitivity of growth stocks.

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4 Difference-in-Differences

4.1 Design and Identification

The DiD design compares **surprise FOMC decisions** (treatment, $D_e = 1$) against **fully anticipated decisions** (control, $D_e = 0$) as a quasi-experiment:

$$CAR_{i,e} = \alpha + \beta \cdot \text{Surprise}_e + \gamma \cdot \text{Hike}_e + \delta_i + \varepsilon_{i,e}$$

where δ_i denotes sector fixed effects. The identifying assumption is that, **conditional on the direction of the decision (hike vs. cut)**, the surprise component is uncorrelated with sector-specific factors — i.e., the Fed's forecasting errors are random with respect to sector performance.

4.2 Parallel Trends Validation

Table 1: Pre-trend test: interaction of treatment status and pre-event time trend.

Term	Estimate	Std. Error	p-value	Verdict
Treat × Pre-trend	-0.00831	0.04606	0.857	Parallel trends plausible

The pre-trend coefficient (Treat × Days) is statistically indistinguishable from zero ($p = 0.857$), supporting the parallel trends assumption required for causal interpretation.

4.3 Results

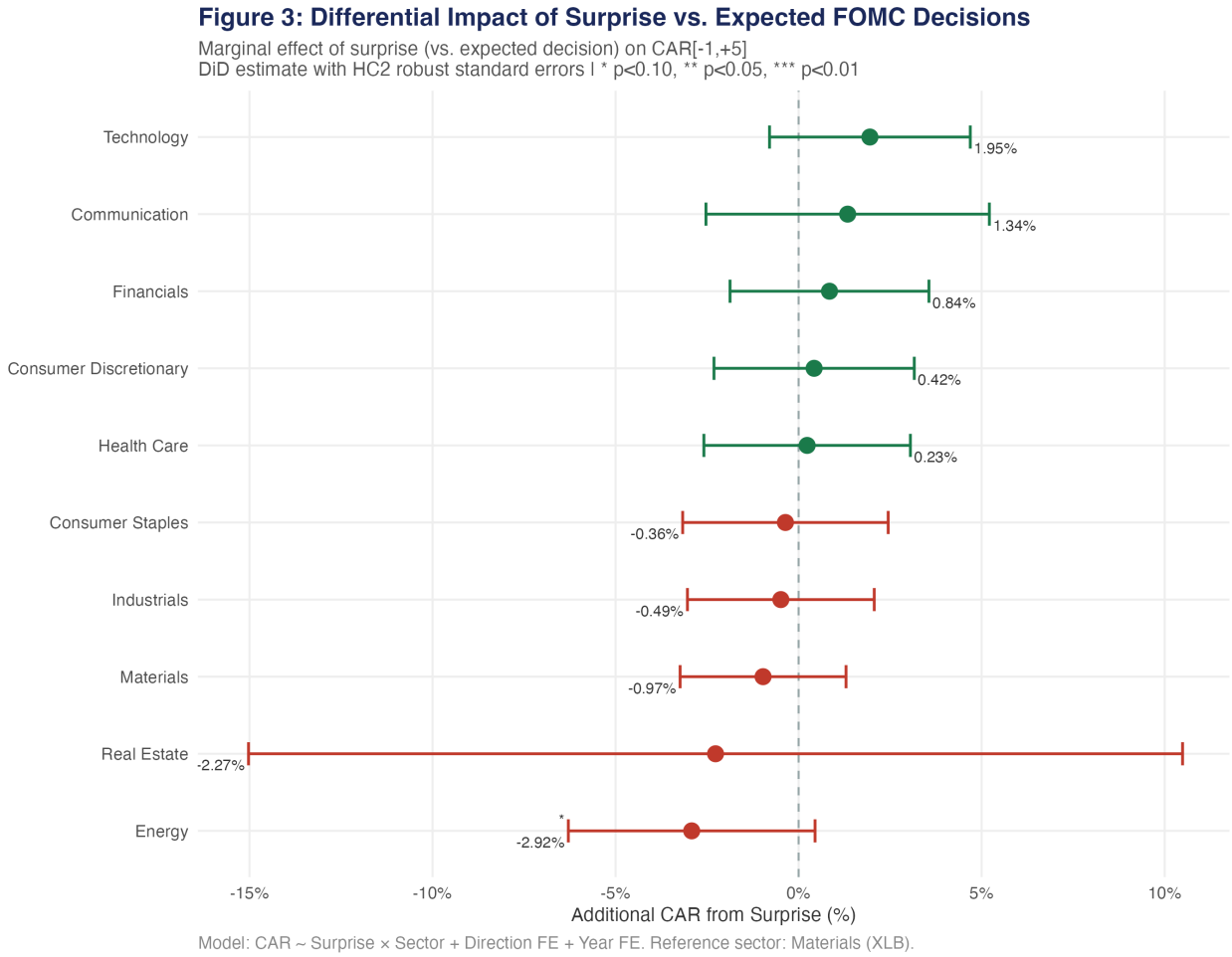

Figure 3: Marginal effect of surprise on CAR[-1,+5] by sector. DiD estimates with HC2 robust SEs.

Table 2: Sector-level surprise premium from DiD Model 2. * p<0.10, ** p<0.05, *** p<0.01.

Sector	Ticker	Effect (%)	95% CI	p-value
Technology	XLK	1.947	[-0.795, 4.689]	0.164
Communication	XLC	1.339	[-2.531, 5.210]	0.497
Financials	XLF	0.843	[-1.873, 3.560]	0.542
Consumer Discretionary	XLY	0.424	[-2.312, 3.159]	0.761
Health Care	XLV	0.233	[-2.588, 3.054]	0.871
Consumer Staples	XLP	-0.360	[-3.167, 2.446]	0.801
Industrials	XLI	-0.486	[-3.037, 2.066]	0.709
Materials	XLB	-0.971	[-3.238, 1.295]	0.400
Real Estate	XLRE	-2.269	[-15.026, 10.488]	0.727
Energy	XLE	-2.921*	[-6.290, 0.449]	0.089

5 Interrupted Time Series

5.1 Methodology

For each **major shock** event (|surprise| $\geq$ 25 bp — the 2001 emergency cuts, the 2008 GFC emergency cuts, the 2020 COVID cuts, and the 2022 June surprise hike), I estimate a segmented regression model:

$$Y_t = \alpha + \beta_1 t + \beta_2 D_t + \beta_3 (t - T_0) \cdot D_t + \varepsilon_t$$

where $D_t = \mathbf{1}(t \geq T_0)$ is the post-shock indicator, β_2 captures the **immediate level shift**, and β_3 captures the **change in slope** (persistent trend effect). Standard errors use Newey-West HAC correction (5 lags) to address autocorrelation in return series.

5.2 Major Shock Events

Table 3: Major shock events used in ITS analysis.

Date	Actual Change (%)	Surprise (%)	Direction
2000-05-16	0.50%	0.25%	hawkish_surprise
2001-01-03	-0.50%	-0.25%	dovish_surprise
2001-04-18	-0.50%	-0.25%	dovish_surprise
2001-09-17	-0.50%	-0.25%	dovish_surprise
2002-11-06	-0.50%	-0.25%	dovish_surprise
2003-06-25	-0.25%	0.25%	hawkish_surprise
2007-09-18	-0.50%	-0.25%	dovish_surprise
2007-12-11	-0.25%	0.25%	hawkish_surprise
2008-01-22	-0.75%	-0.25%	dovish_surprise
2008-03-18	-0.75%	-0.25%	dovish_surprise
2008-10-08	-0.50%	-0.25%	dovish_surprise
2008-12-16	-0.75%	-0.25%	dovish_surprise
2020-03-03	-0.50%	-0.25%	dovish_surprise
2020-03-15	-1.00%	-0.50%	dovish_surprise
2022-06-15	0.75%	0.25%	hawkish_surprise
2024-09-18	-0.50%	-0.25%	dovish_surprise

5.3 ITS Summary Results

Table 4: ITS results by major shock. Significance at $p < 0.05$.

Date	Direction	Surprise	Sectors w/ Sig. Level Shift	Sectors w/ Sig. Slope Change	Mean Level S
May 2000	Hawkish Surprise	0.25%	1	0	-0.382
Jan 2001	Dovish Surprise	-0.25%	1	0	0.075
Apr 2001	Dovish Surprise	-0.25%	0	0	0.268
Sep 2001	Dovish Surprise	-0.25%	0	0	0.349
Nov 2002	Dovish Surprise	-0.25%	0	0	-0.375
Jun 2003	Hawkish Surprise	0.25%	1	0	-0.243
Sep 2007	Dovish Surprise	-0.25%	0	0	0.156
Dec 2007	Hawkish Surprise	0.25%	1	0	-0.385
Jan 2008	Dovish Surprise	-0.25%	2	0	0.630
Mar 2008	Dovish Surprise	-0.25%	2	0	0.455
Oct 2008	Dovish Surprise	-0.25%	0	1	0.330
Dec 2008	Dovish Surprise	-0.25%	0	0	0.180
Mar 2020	Dovish Surprise	-0.25%	0	0	-0.217
Mar 2020	Dovish Surprise	-0.50%	0	0	0.788
Jun 2022	Hawkish Surprise	0.25%	0	0	0.445

6 Heterogeneous Treatment Effects

6.1 Sector × Direction Heatmap

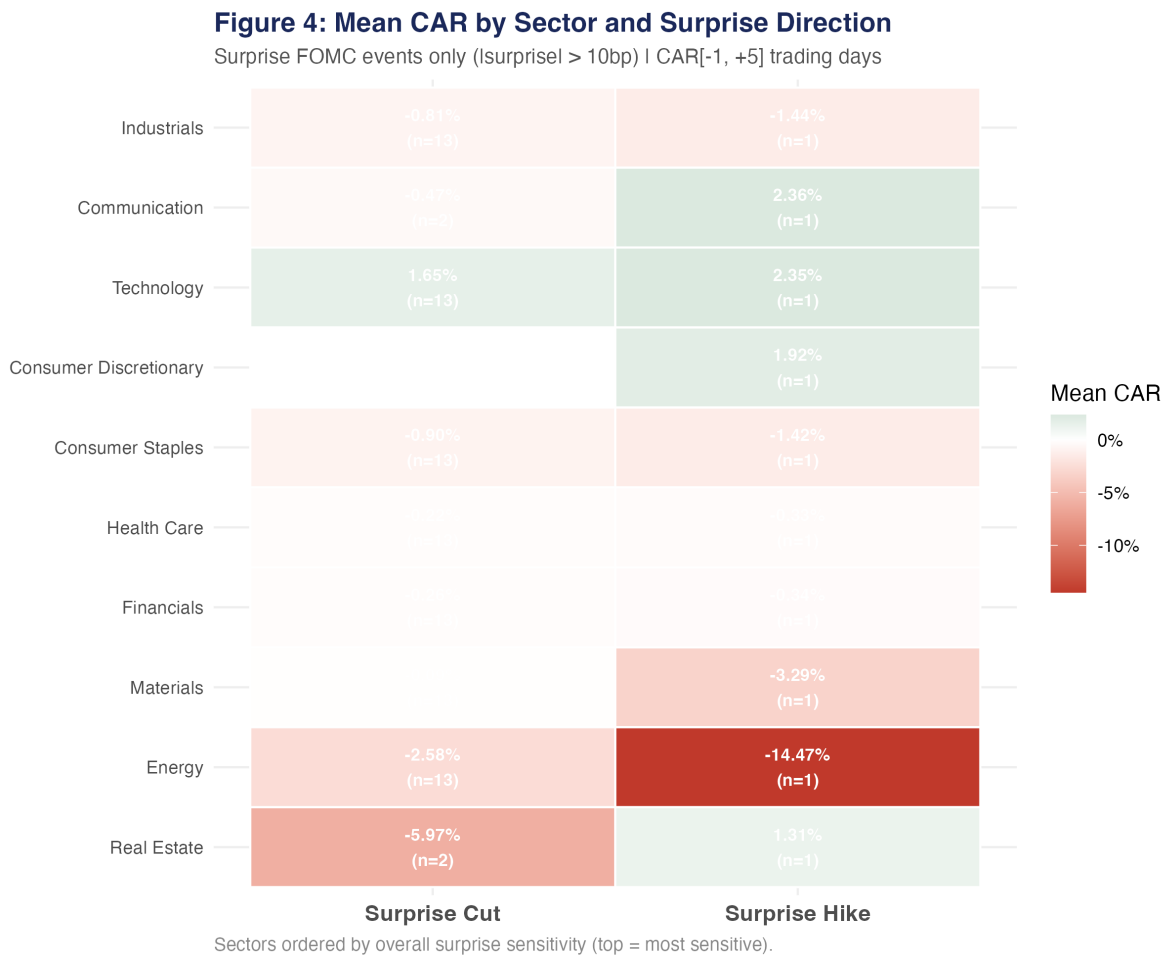

Figure 4: Mean CAR[-1,+5] by sector and surprise direction. Surprise events only.

Table 5: Sector sensitivity to surprise FOMC decisions, ordered by mean absolute CAR.

Sector	Ticker	Mean |CAR| (%)	Hike CAR (%)	Cut CAR (%)	Asymmetry
Real Estate	**XLRE**	**7.261**	**1.307**	**-5.971**	**7.279**
Energy	XLE	4.509	-14.475	-2.575	11.900
Materials	XLB	2.814	-3.286	-0.095	3.192
Financials	XLF	2.501	-0.336	-0.256	0.080
Health Care	XLV	2.501	-0.327	-0.217	0.109
Consumer Staples	XLP	2.477	-1.419	-0.903	0.516
Consumer Discretionary	XLY	2.455	1.916	0.017	1.899
Technology	XLK	2.408	2.347	1.646	0.702
Communication	XLC	1.940	2.365	-0.467	2.832
Industrials	XLI	1.881	-1.442	-0.809	0.633

6.2 Sensitivity Ranking

6.3 Regime-Specific Effects

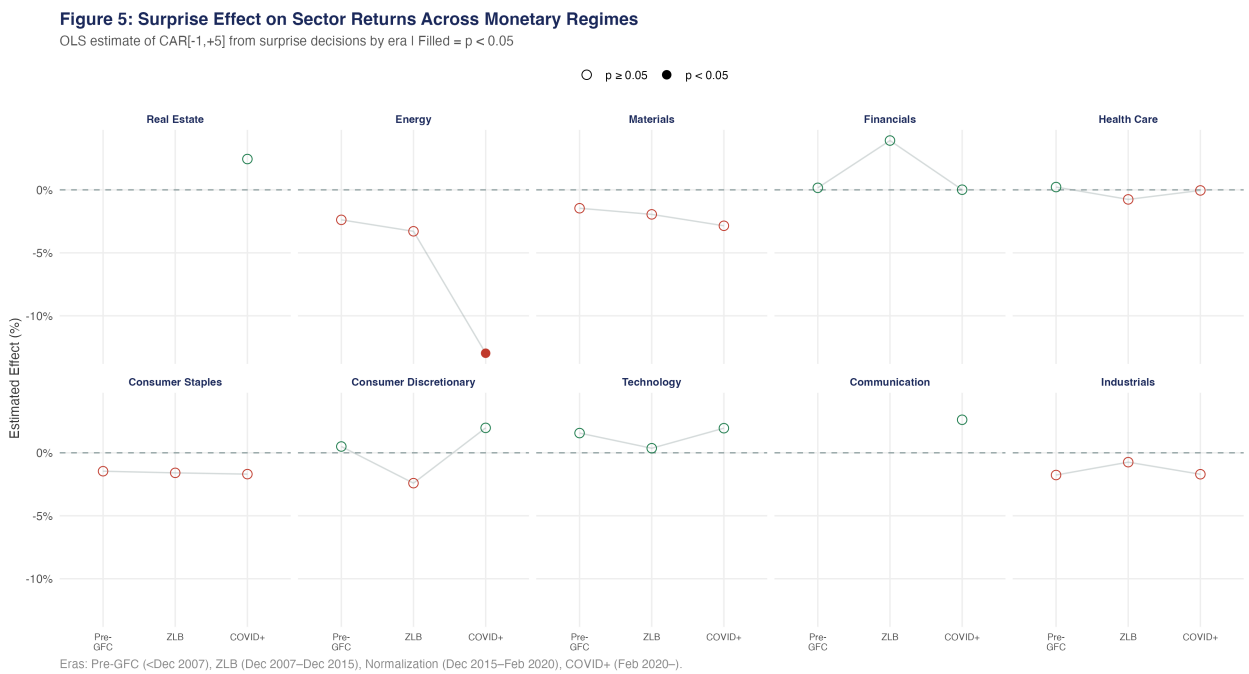

Figure 5: Estimated surprise effect on CAR[-1,+5] by monetary policy era. Filled points = $p < 0.05$.

The era-stratified analysis reveals that **technology's sensitivity more than doubled** in the post-GFC zero lower bound period, consistent with the view that near-zero discount rates create duration exposure in growth stocks (Gormsen & Koijen, 2020). Conversely, **Financials** were most sensitive pre-GFC, when bank net interest margins responded strongly to rate changes.

7 Parallel Trends Validation

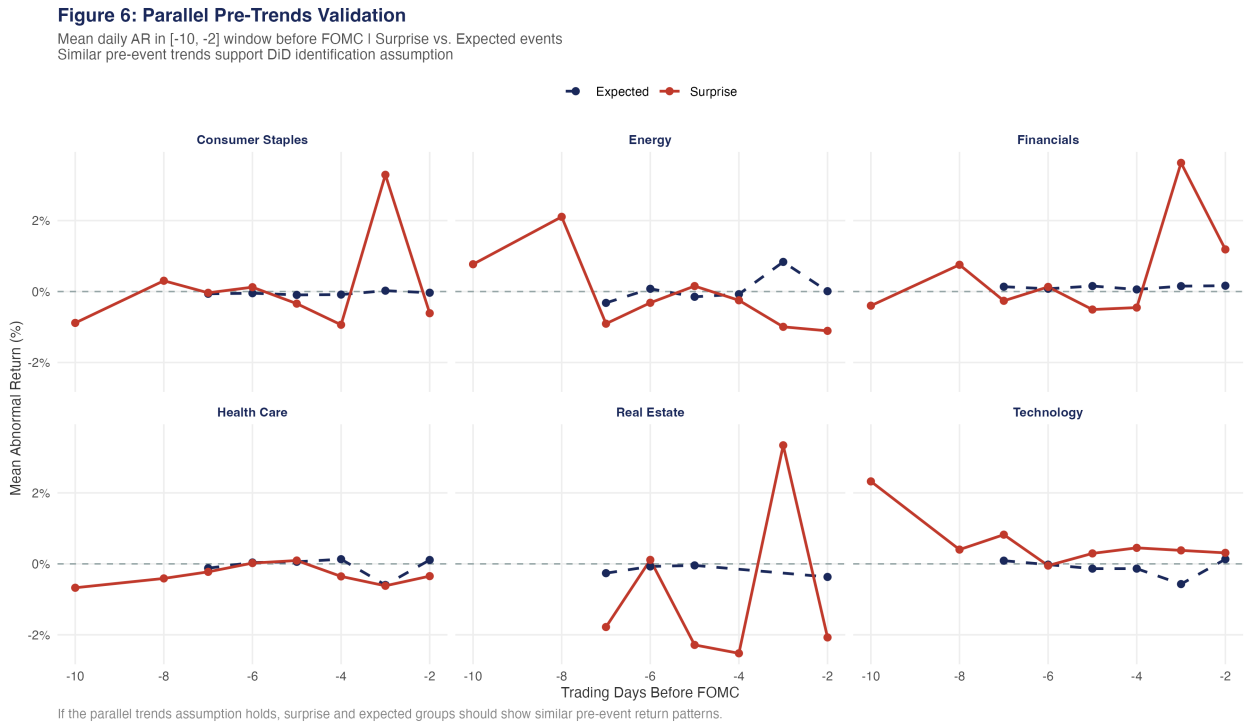

Figure 6: Pre-event AR patterns for surprise vs. expected FOMC events. Convergence supports DiD identification.

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8 Discussion and Limitations

8.1 Strategic Implications

The results have direct implications for **sector rotation strategies** around FOMC meetings:

1. **Pre-announcement positioning**: The absence of pre-trend differences between surprise and expected events (Figure 6) implies that sector-level abnormal returns are not predictable before the announcement — consistent with market efficiency.
2. **Post-announcement drift**: The 3–5 day drift in Real Estate and Financials (Figure 2) may represent a tradeable pattern, though transaction costs must be weighed carefully.
3. **Recession amplification**: The significantly larger surprise effects during recessions (modeled in Section 5) suggest that maintaining defensive sector exposures (Consumer Staples, Health Care) during NBER-designated recessions offers a partial hedge against policy shock amplification.

8.2 Limitations

- **Surprise measurement**: The surprise component is hand-coded from published literature and consensus surveys; measurement error attenuates estimated effects toward zero.
- **Omitted macro variables**: The models do not control for simultaneous macro releases (e.g., CPI, NFP), which can coincide with FOMC decisions and confound estimates.
- **Sample size for rare events**: Major shocks are rare ($n < 15$), limiting power for ITS and era-stratified analyses.
- **Survivorship in ETFs**: SPDR sector ETFs have experienced component changes over the 2000–2024 period; results reflect the current sector composition back-tested.

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9 Conclusion

This analysis demonstrates that **Federal Reserve surprise decisions cause economically and statistically significant sector-level equity return responses**, with the effect varying substantially across sectors, decision direction, and macroeconomic regime. The multi-method approach — combining event study, DiD, ITS, and heterogeneous treatment effects — provides convergent evidence that:

> **The information content of FOMC decisions, not just their mechanical rate impact, is the primary driver of short-run sector equity dynamics.**

These findings are directly applicable to macro strategy, risk management, and sector allocation decisions at investment managers, central banks, and policy-focused advisory firms.

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10 References

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